

File permissions in Linux

Project description

The research team at my organization needs to ensure that the users on their team are authorized with the appropriate permissions. For that, the research team examines the existing file permissions within the **projects** directory and determines whether there are any unauthorized permissions to access the files. If there is any unauthorized permission, then the team modify the file permissions to authorize the appropriate users and remove any unauthorized access. Checking and updating the file permission helps to keep the system secure. To complete this task, I performed the following tasks:

Check file and directory details

```
researcher2@b2c5fb1e0a67:~$ pwd
/home/researcher2
researcher2@b2c5fb1e0a67:~$ cd projects
researcher2@b2c5fb1e0a67:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 18 12:49 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 12:49 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 12:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_t.txt
researcher2@b2c5fb1e0a67:~/projects$ ls -la
.  .project_x.txt  project_k.txt  project_r.txt
.. drafts        project_m.txt  project_t.txt
researcher2@b2c5fb1e0a67:~/projects$
```

Here, I explored the permissions of the projects directory and file it contains. First, the current working directory was **/home/researcher2** and to check for permission of projects directory, I navigate to the projects directory using the command **cd projects**. Now, we list the contents and permissions of the projects directory using **ls -l**. The permissions of files of the projects directory is shown above by looking at the 10-character string. Let's take an example from the screenshot above: **-rw-rw-rw-**, here it indicates that the file **project_k.txt** is a regular with read and write permission for user, groups and other but not executable permission for user, groups and others.

To check whether there is an existing hidden file which consists of period (.) before the file name, type **ls -la** command to check for hidden files and to check the permission for the hidden files type **ls -la** command.

Describe the permissions string

```
researcher2@b2c5fb1e0a67:~$ cd projects
researcher2@b2c5fb1e0a67:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 18 12:49 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 12:49 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 12:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_t.txt
```

From the screenshot above, we can see the permissions of the files and directory in the projects directory. On the left side, we can see a 10-character string like: **drwxrwxrwx**, which represents the permissions granted for the users, groups and others. Let's take a review :

1. The first character represents the file type. Here d indicates a directory and if the hyphen is there, then it is a regular file.
2. The 2nd to 4th character(rwx) represents the read(r),write(w) and execute(x) permissions for the user. When one of the characters is hyphen then it is not granted this permission is not granted to the user.
3. The 5nd to 7th character(rwx) represents the read(r),write(w) and execute(x) permissions for the group.When one of the characters is hyphen then it is not granted this permission is not granted to the group.
4. The 8nd to 10th character(rwx) represents the read(r),write(w) and execute(x) permissions for the others.When one of the characters is hyphen then it is not granted this permission is not granted to the others.

By looking at the 10-character string for each file or directory, we can determine what all permissions are granted for the file.

Change file permissions

```
researcher2@b2c5fb1e0a67:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 18 12:49 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 12:49 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 12:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_t.txt

researcher2@b2c5fb1e0a67:~/projects$ chmod o-w project_k.txt
researcher2@b2c5fb1e0a67:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 18 12:49 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 12:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_t.txt
```

Here, we determine whether any of the files have incorrect permissions and then change the permissions as needed by the organization. Here, the organization restricts the others from having write permissions on any files.

As you can see there is a write permission for others in the file named **project_k.txt**. now we have to modify the file permission so that **project_k.txt** doesn't grant write permission to others. Type **chmod o-w project_k.txt** command where o represents others and w as write and - indicates that to take away the write permission from others. Now type **ls -l** to ensure that there is no write permission for others in the file.

Change file permissions on a hidden file

```
researcher2@caefea90b345:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 13:56 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 14:42 ..
-rw--w---- 1 researcher2 research_team  46 Jun 18 13:56 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 18 13:56 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 13:56 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 13:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 13:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 13:56 project_t.txt
researcher2@caefea90b345:~/projects$ chmod u-w,g-w .project_x.txt
researcher2@caefea90b345:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 13:56 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 14:42 ..
-r----- 1 researcher2 research_team  46 Jun 18 13:56 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 18 13:56 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 13:56 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 13:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 13:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 13:56 project_t.txt
researcher2@caefea90b345:~/projects$ chmod g+r .project_x.txt
researcher2@caefea90b345:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 13:56 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 14:42 ..
-r--r----- 1 researcher2 research_team  46 Jun 18 13:56 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 18 13:56 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 13:56 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 13:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 13:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 13:56 project_t.txt
researcher2@caefea90b345:~/projects$
```

The research team has archived the hidden file **.project_x.txt** by typing **ls -la** command. This file should not have write permissions for anyone, but the user and group should be able to read the file. But we can see that **.project_x.txt** has write permissions on user and group and group doesn't have read permission. So we need to modify the permissions of **.project_x.txt**. Type **chmod u-w,g-w .project_x.txt** to take away the write permission from the user and groups of **.project_x.txt** file. Type **chmod g+r .project_x.txt** command to add read permission for the groups. Otherwise, we can write it altogether like this: **chmod u-w,g-w,g+r .project_x.txt**. type **ls -la** and you can see that there is read permission for groups and no write permission for the user and groups.

Change directory permissions

```
researcher2@b2c5fb1e0a67:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 18 12:49 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 18 12:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_t.txt
researcher2@b2c5fb1e0a67:~/projects$ chmod g-x drafts
researcher2@b2c5fb1e0a67:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Jun 18 12:49 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 18 12:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:49 project_t.txt
researcher2@b2c5fb1e0a67:~/projects$
```

The files and directories in the projects directory belong to the **researcher2** user. Only **researcher2** should be allowed to access the **drafts** directory and its contents. The research team checks the permission of the **drafts** directory inside the project directory. We can see that there is an execute permission for groups in the **drafts** directory and it is mentioned that only researcher2 can access the **drafts** directory. We need to modify the permissions for **drafts** directory. Type `chmod g-x drafts` to take away execute(x) permission from the **drafts** directory. To confirm that, type `ls -l` and we can see that there are no execute permissions for groups.

Summary

By performing these all tasks, I changed multiple file permissions and ensured that there are no unauthorized file permissions within the projects directory. The first step is checking by using `ls -l` or `ls -la` command to check for file permissions. Then, we use the **chmod** command multiple times to modify the file permissions within the **projects** directory.